PART 1

Where the project stands

How the project got from a hand-wired armature rig to a Follow Path framework, what the add-on does today, and what the code audit found.

1.1 From the legacy rig to Track It
1.2 What the add-on can do today
1.3 How the current code is wired
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1.1 From the legacy rig to Track It

The original train rigs were built years ago through trial and error, and they worked. But every train was hard-wired to one track by name, and the motion came from a chain of indirect constraints:

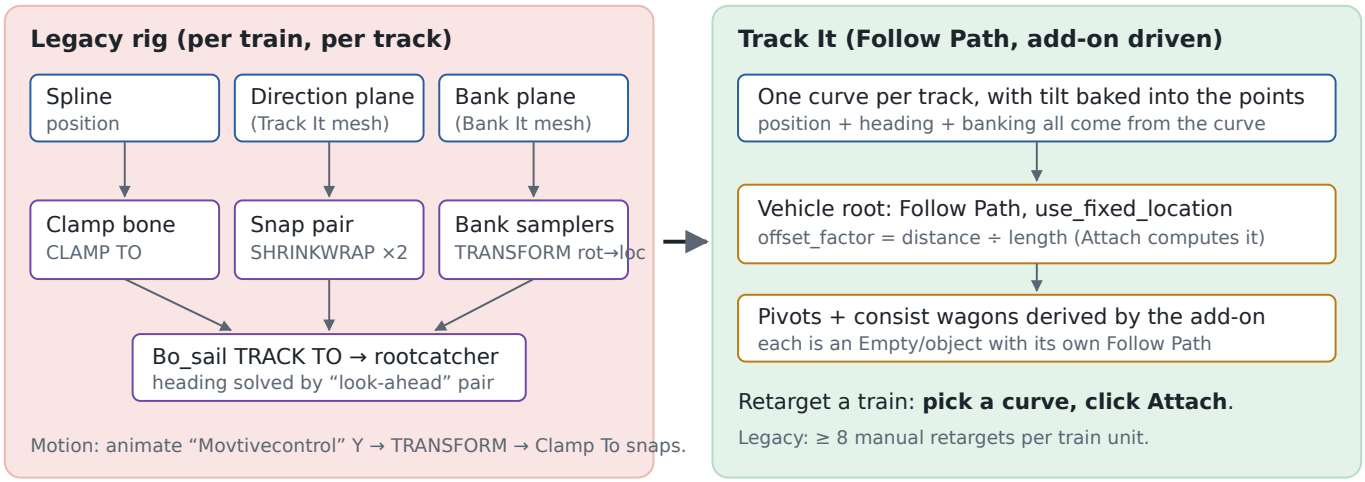


Figure 1.1 Left: the legacy rig reconstructed in session 1 (the “running engine 1” locomotive). Right: the architecture that replaced it.

Milestones so far

When	What changed
May 2026	Legacy rig analysed. Wish list set: attach where the object stands, fewer retargets, track switching.
4–5 Sep	Follow Path found to give position, heading and banking from one curve. Attach operator: closest point → arc-length offset_factor, then zero the local location.
7 Sep	Nearby Curves list (nearest point on curve). Drag Along Curve via proxy Empty, Drag Power, keyframe toggle. Detach mutes instead of deleting. Rotation reset on Attach.

When	What changed
8–10 Sep	Status text, Switching panel, proximity detection with Enter/Exit range, switch keyframing, future-keyframe cleanup. Drag Slider replaces Mouse Drag as the main control. Three modal-transform crashes lead to a standing rule.
11–14 Sep	Vehicle panel with pivot list, auto-created pivot Empties on their own Follow Path, Pivot Data stub. Mute/enabled replaced by influence. Live switching, markers, EXTREME/BREAKDOWN keyframe types, cascade Remove Switch, auto-switch reconstruction.
After v8	Consist panel with wagons, cumulative route distance, route-history log, junction solving, wagons deciding forks themselves, stale-history pruning, Fix Train, Consist Debug box. *(Found in the latest source; not described in handoff v8.)*

1.2 What the add-on can do today

Area	Feature	Status
Track tools	“Track It” / “Bank It” mesh generators (leftovers from the legacy rig)	Legacy
Attach	Attach selected at current world position, curve picker, Nearby Curves list sorted by nearest point	Built
Attach	Detach bakes the pose and sets influence to 0 (constraint and keyframes kept)	Built
Drive	Drag Slider (centre 0.5, springs back) + Drag Power	Built
Drive	Mouse Drag (proxy Empty + G)	Deprecated
Drive	Drag Pad (2D N-panel pad, crash-safe Mouse Drag replacement)	Prototype, not in file
Keyframes	Keyframe all held constraints (offset + influence), context label, Remove Switch cascade with count dialog	Built
Switching	Candidate discovery by proximity, Enter/Exit hysteresis, Switch button, HUD notice, marker + EXTREME key	Built
Switching	Live switching (recorded fork, or unique continuation at a curve end), behind a checkbox	Behind toggle
Switching	Auto-switch reconstruction (BREAKDOWN key from stashed crossing)	Built
Vehicle	Pivot list: auto Empties, offset in metres, same-curve Follow Path	Built
Vehicle	Pivots crossing onto a different curve than the root	Not built
Vehicle	Rig bone / object binding (Pivot Data fields)	Stored, inert

Area	Feature	Status
Consist	Wagon list, coupling distances, route-distance history, wagons crossing forks and deciding forks when leading	Prototype
Consist	Vehicle library, coupling/uncoupling, driving from any car	Not built
Wheels / rig	Auto wheel rotation, Direct mode, slip, sway, hunting, suspension	Concept only
Assets	Vehicle definition, asset library, validation	Concept only

1.3 How the current code is wired

The add-on is one 3,143-line file. It has three kinds of moving parts: **operators** the user clicks, **handlers** Blender calls on every update or frame change, and **data** stored on objects and the scene. The handlers do most of the work.

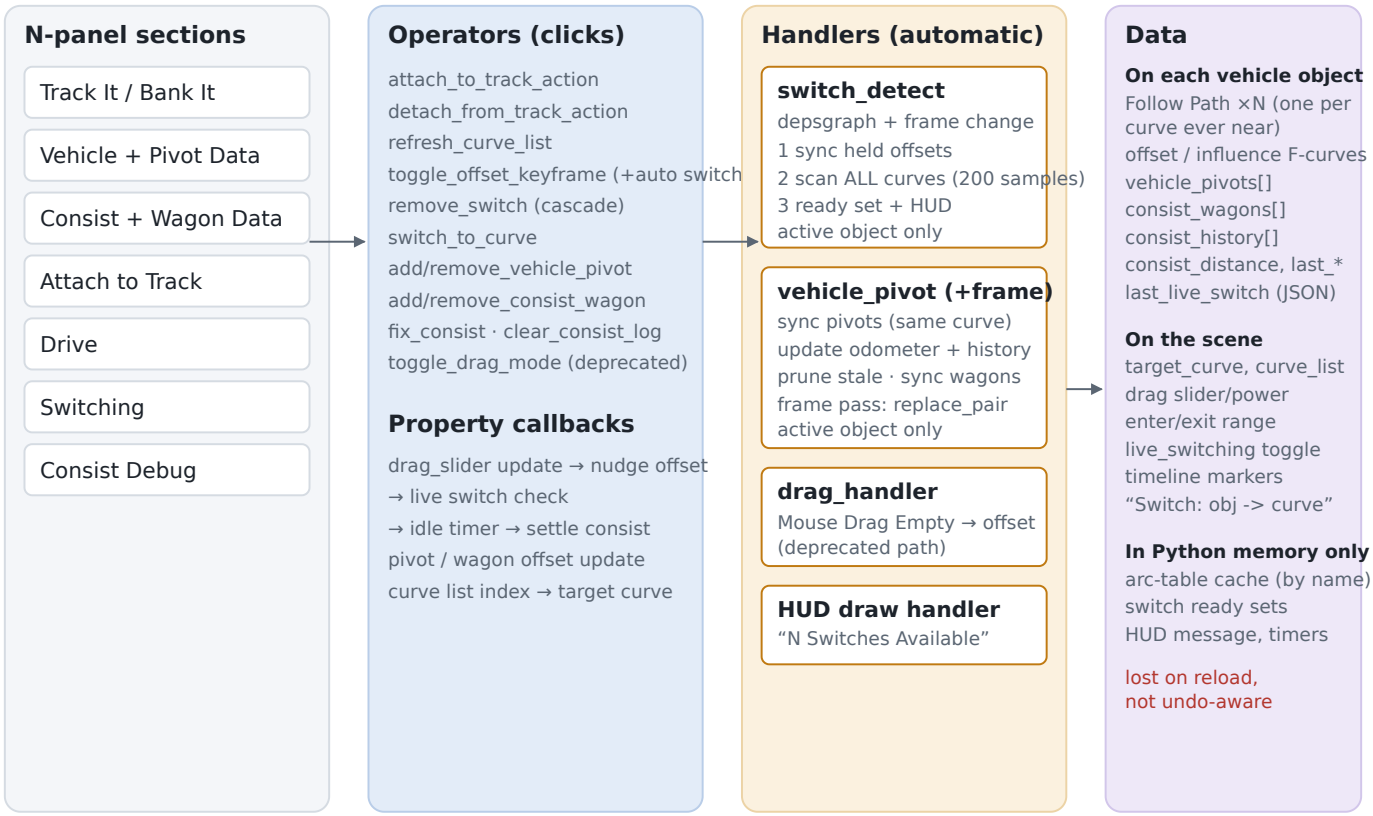


Figure 1.2 Wiring map of `Track it compiled 1.py`. Both main handlers act only on the *active* object, which is behind the render and playback issues listed in 1.5.

1.4 Handoff v8 vs the latest source

RECONCILING THE TWO SOURCES

- The latest source **goes beyond handoff v8**. It contains a full consist prototype: wagons, `TrackIt_ConsistHistoryEntry`, a cumulative route-distance odometer, `_track_it_find_junction`, `_track_it_resolve_route_position`, stale-history pruning, Fix Train, and the Consist Debug box. That is v8's Section 8.2/8.3 "theory" partly built for *wagons*, but not yet for *pivots*.
- The **Drag Pad** prototype v8 describes (`_track_it_draw_drag_pad`, `OT_DragPadGrab_Action`) is **not in this file**. It is probably in the backed-up fork. Worth recovering before Phase 2 (Part 11).
- `bl_info` still says version **(1, 32)** and Blender **(3, 0, 0)**. The code uses the layered-Action API (`action.layers → strips → channelbag`), which only exists in recent Blender (the project runs 5.1). The minimum version should be raised so older Blenders refuse to load it cleanly.
- A **v9 handoff** should record the consist work. This guide can stand in until then.

1.5 Code audit: issues found

Line numbers refer to the latest `Track it compiled 1.py`. Severity: High gives wrong results or breaks renders, Med is fragile or slow, Low is tidy-up.

#	Sev.	Issue	Why it matters / fix direction
1	High	**Syncing only happens for the active object.** Both handlers start with `view_layer.objects.active` (L1171, L2361).	Pivots, wagons and held-offset sync freeze whenever a different object is selected: during playback, during a render, or when a second train exists. Fix: evaluate every registered vehicle, not the selection (Part 7, render-safe evaluation).
2	High	**Arc-table cache never invalidates.** `_track_it_switch_arc_cache` is keyed by curve *name* and only cleared on unregister (L1036–1046, L3111).	Edit, extend or move a curve after it has been used and every offset computed against it is silently wrong until the add-on reloads. Renaming a curve can also hit a stale entry. Fix: key by object pointer plus a hash of points, handles, tilt, `matrix_world` and resolution.
3	High	**Pivots still clamp to the root's curve** (L1731). The consist code already solved crossing for wagons but pivots don't use it.	This is v8's §3.3 gap: a pivot past a curve end snaps to the boundary and the wheelbase breaks. Fix: pivots resolve through the same route resolver as wagons (Part 6).

#	Sev.	Issue	Why it matters / fix direction
4	High?	**Wagons clone the root with `root.copy()`** (L2554). Object-level collection properties (`track_it_vehicle_pivots`, `track_it_consist_wagons`, history) are ID properties and are very likely copied too.	A wagon may inherit a wagon list that points at itself and the root's pivot Empties. If the wagon ever becomes active, the handler would sync that inherited list and fight the real root. *Verify in Blender*, then clear those collections on the clone.
5	Med	**Per-tick cost grows with the scene.** Job 2 scans *every* curve at 200 samples on every depsgraph update. `closest_offset_factor` is a 5,000-point brute-force loop.	Fine for a test scene, slow on a real layout. Fix: a precomputed Track Network (Part 4) plus a spatial grid or KD-tree (`mathutils.kdtree`) per curve.
6	Med	**Handlers write RNA inside `depsgraph_update_post`.**	Every write triggers another update pass. Guards (skip if unchanged) keep this finite, but it is a feedback loop. Prefer `frame_change_pre` for animation-driven work and do UI-driven work in the operator or callback itself.
7	Med	**Two history systems.** Keyframe mining (`_find_recorded_switch_point`) for the root, and the odometer log for wagons.	They can disagree. Keyframe mining was already contaminated twice by leftover test keys, and the odometer needs drift heuristics (`replace_pair`, DEDUP_EPS). Fix: one Route Plan (Part 5).
8	Med	**Identity by name strings.** Ready sets store `curve.name`, markers encode `obj.name`, constraints are named `FollowPath_{curve}`.	Renaming a vehicle orphans its switch markers, and renaming a curve breaks constraint lookup names. Fix: a stable UUID custom property on every managed object; match by pointer.
9	Med	**`build_arc_table` ignores cyclic splines** and spreads samples evenly per segment regardless of segment length.	Closed loops (roundy-round layouts) are missing the closing segment. Long segments get fewer samples per metre. Fix: sample the *evaluated* curve (depsgraph polyline Blender itself follows) including the cyclic segment.
10	Med	**Python-only state isn't undo-aware** (ready sets, caches).	After Ctrl+Z the Switching panel can show stale candidates until the next move. Fix: rebuild on `undo_post` / `load_post`.
11	Low	The Drag Slider registers a new timer on every update (L473).	Harmless but wasteful. Check `bpy.app.timers.is_registered` first.
12	Low	Mute/enabled writes still sprinkled through the code after the influence migration.	Add a one-time “Migrate constraints” operator, then delete the dual-state code.
13	Low	“Track It” / “Bank It” operators generate legacy shrinkwrap planes with `bpy.ops` chains.	Nothing in the new architecture needs them. Replace with a Track Builder (Part 4).
14	Low	Single file, globals, no tests.	Split into a package (Part 3 §3.5) and add headless Blender tests (Part 11).

1.6 What “finished” means

Pulling the goals from every session together, the add-on is complete when an artist can:

1. **Build or import track** quickly, with banking and switches, and have the add-on understand how the pieces connect.
2. **Turn any model into a vehicle** (mine cart, subway car, 2-4-2 steam locomotive) through a guided set-up, and save it to a library.
3. **Couple vehicles into a train**, drag it anywhere including backwards through switches, and have every car and every bogie stay on the correct rails.
4. **Animate it** with ordinary keyframes and easing, with wheels, rods and secondary motion following automatically, or take manual control of any wheel for slip or slide.
5. **Render it reliably**, on a farm or in the background, and get the same result every time.

The rest of this guide is the plan for getting there.